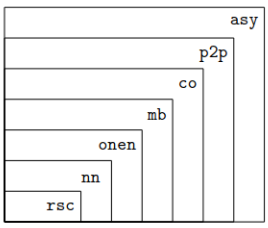
Notes

A Partial Order View of Message-Passing Communication Models (POPL23)

$\mathbb{P}$: set of processes/partecipants $(p, q, r, ...)$

$\mathbb{M}$: set of messags $(m1, m2, m3, ...)$



(a) The hierarchy of MSC classes.

	Weakly sync	Weakly k-sync	∃k bounded	∀k bounded
asy	unbounded STW	✓	✓	✓
p2p	✗ [1]	✓ [1]	✓ [1]	✓ [1]
co	✗	✓	✓	✓
mb	✓ [1]	✓ [1]	✓ [1]	✓ [1]
onen	✓	✓	✓	✓
nn	✓	✓	✓	✓

(b) (Un)decidability results for the synchronizability problems, [1] indicates that the result was shown in [3].

Figure 2: Main contributions

Definition of Message Sequence Chart (MSC)

An MSC over $\mathbb{P}$ and $\mathbb{M}$ is a tuple $M = (\mathrm{E}, \rightarrow, \lhd, \lambda)$ where E is finite (possibly empty) set of events, $\lambda : \mathrm{E} \rightarrow \Sigma$ is a labelling function that associates an action to each event, and $\rightarrow, \lhd$ are binart relations on E that satisfy the following three conditions. For $p \in \mathbb{P}$, let $\mathrm{E}_p = \{e \in \mathrm{E} \mid \lambda(e) \in \Sigma_p\}$ be the set of vents that are executed by p.

1. The process relation $\rightarrow \subseteq \mathrm{E} \times \mathrm{E}$ relates an event to its immediate successor on the same process: $\rightarrow = \bigcup_{p \in \mathbb{P}} \underset{p}{\rightarrow}$ for some relations $\underset{p}{\rightarrow} \subseteq \mathrm{E}_p \times \mathrm{E}_p$ such that $\underset{p}{\rightarrow}$ is the direct successor relation of a total order on E_p.
2. The message relation $\lhd \subseteq \mathrm{E} \times \mathrm{E}$ relates pair of matching send/receive events:

   2a. for every pair $(e, f) \in \lhd$, there are two processes p, q and a message m such that $\lambda(f) = \mathrm{send}(p, q, m)$and $\lambda(f) = \mathrm{rec}(p, q, m)$.

   2b. for all $f \in \mathrm{E}$, with $\lambda(f) = \mathrm{rec}(p, q, m)$, there is exactly one $e \in \mathrm{E}$ such that $e \lhd \mathrm{E}$.

   2c. for all $e \in \mathrm{E}$ such that $\lambda(f) = \mathrm{send}(p, q, m)$, there is at most one $f \in \mathrm{E}$ such that $e \lhd f$.
3. The happens-before relation $\underset{\mathrm{hb}}{\leq}$, defined by $(\rightarrow \cup \lhd)^*$, is a partial order on E.

Fully async

In the fully asynchronous communication model (`asy`), messages can be received at any time once they have been sent, and send events are non-blocking. It can be modeled as a bag where all messages are stored and retrieved by processes when necessary. It is also referred to as NON-FIFO. The definition is the same as MSC.

Peer-to-peer

In the peer-to-peer (`p2p`) communication model, any two messages sent from one process to another are always received in the same order as they are sent. This is usually implemented by processes pairwise connected with FIFO channels. Alternative names are FIFO 1–1 or simply FIFO.

A p2p-MSC is an MSC $M = (E, \rightarrow, \lhd, \lambda)$ where, for any two send events s and s' such that $\lambda(s) \in \mathrm{send}(p, q, _)$, $\lambda(s') \in \mathrm{send}(p, q, _)$, and $s \overset{+}{\rightarrow} s'$, one of the following holds:

- either $s, s' \in \mathrm{matched}(M)$ with $s \lhd r$ and $s' \lhd r'$ and $r \overset{+}{\rightarrow} r'$,

- or $s' \in \text{unmatched}(M)$.

Note that we cannot have two messages $m1$ and $m2$, both sent by p to q, in that order, such that $m1$ is unmatched and $m2$ is matched; unmatched message $m1$ excludes the reception of any later message.

Causally ordered

In the causally ordered (`co`) communication model, messages are delivered to a process according to the causality of their emissions. In other words, if there are two messages $m1$ and $m2$ with the same recipient, such that there exists a causal path from $m1$ to $m2$, then $m1$ must be received before $m2$. Causal ordering was introduced by Lamport with the name "happened before" order.

An MSC $M = (E, \rightarrow, \lhd, \lambda)$ is causally ordered if, for any two send s and s', such that $\lambda(s) \in \text{send}(_, q, _)$, $\lambda(s') \in \text{send}(_, q, _)$, and $s \underset{\text{hb}}{\leq} s'$:

- either $s, s' \in \text{matched}(M)$ and $r \overset{*}{\rightarrow} r'$, with r and r' receive events such that $s \lhd r$ and $s' \lhd r'$.
- or $s' \in \text{unmatched}(M)$.

Note that in a `co`-MSC we cannot have two send events s and s' addressed to the same process, such that s is unmatched, s' is matched, and $s \underset{\text{hb}}{\leq} s'$.

Mailbox

FIFO 1-n

FIFO n-n

RSC

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Definition of conflict graph

Intuitively, we have a dependency whenever two messages have a process in common.

The conflict graph `CG(e)` of a sequence of action $e = a_1 ... a_n$ is the labeled graph $\left(V, \{\overset{XY}{\rightarrow}\}_{X,Y \in \{R,S\}}\right)$ where V is the set of message exchanges of e, and for all $X, Y \in \{S, R\}$, for all $v, v' \in V$, there is a XY dependency edge $v \overset{XY}{\rightarrow} v'$ between v and v' if there are $i < j$ such that $\{a_i\} = v \cap X$, $\{a_j\} = v' \cap Y$, and $\text{proc}_{X(v)} = \text{proc}_{Y(v')}$.

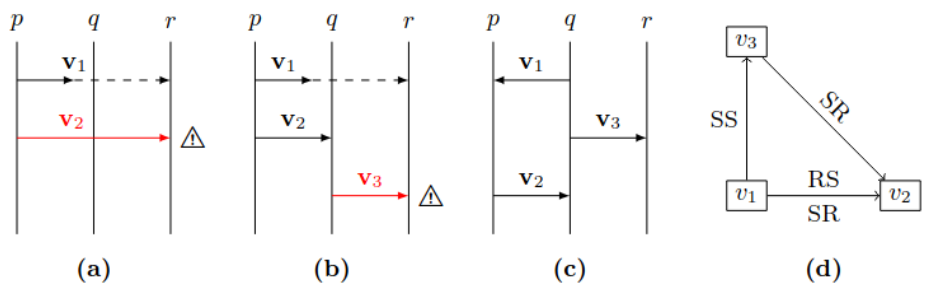

Fig. 1: (a) and (b): two MSCs that violate causal delivery. (c) and (d): an MSC and its conflict graph

Definition of k-synchronous

k denotes a given integer $k \geq 1$. A k-exchange denotes a sequence of action starting with at most k sends and followed by at most k receives matching some of the sends. An MSC is k-synchronous if there exists a linearisation that is breakable into a sequence of k-exchanges sucj that a message sent

during a k-exchange cannot be received during a subsequent one: either it is received during the same k-exchange, or it remains orphan forever.

An MSC `msc` is k-synchronous if:
1. there exists a linearisation of m $e = e_1 \cdot e_2 ... e_n$ where for all $i \in [1...n]$, $e_i \in S^{\leq k} \cdot R^{\leq k}$,
2. m satisfies causal delivery,
3. for all j, j' such that $a_j \mid - \mid a_{j'}$ holds in e, $a_j \mid - \mid a_{j'}$ holds in some e_i.

An execution e is k-synchronizable if $m(e)$ is k-synchronous.

We write $sTr_k(\mathfrak{S})$ to denote the set $\{m(e) \mid e \in asEx(\mathfrak{S})$ and $m(e)$ is k-synchronous$\}$.

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Definition of borderline violation

Let $\bigwedge_{\mathbb{P},\mathbb{I},\mathbb{V}}$ be an alphabet of communications. A borderline violation is an execution $e = e' \cdot \iota ?^p \mathrm{v}$ over $\bigwedge_{\mathbb{P},\mathbb{I},\mathbb{V}}$ such that e' is RSC, and e is not causally equivalent to an RSC execution.

Random notes

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